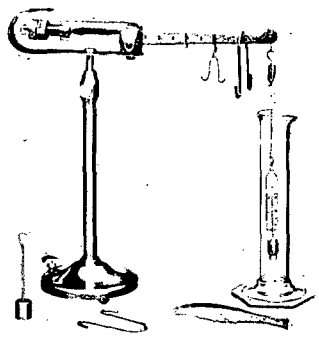
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
This military standard is approved by the Department of the and shall be used by that activity. All other military activities are required to employ this standard where suitable.

USER ACTIVITIES: ARMY - AV, ER, ME
NAVY - CG, OS, YD

REVIEWER ACTIVITIES: ARMY - AR

FED. SUP CLASS
6630



PART NUMBER	NOMENCLATURE
MS36537-1	Specific Gravity Determination Balance
MS36537-2	Thermometer Plummet (Replaceable Component Part)

PLUMMET DATA		THERMOMETER DATA			
SP GR RANGE	DISPLACEMENT AT 15°C	RANGE	ACCURACY RATING	WEIGHT UNITS PER SET (GRAMS)	BALANCE SENSITIVITY
0.7 to 2	5 mL	10° to 30°C	1°C	5.0, 0.5, 0.05, 0.005 2 each	0.0001 g

MATERIAL: (A) Hydrometer jar shall be of borosilicate glass in accordance with ASTM E 438, Type I, Class A.

Counterpoise weight and beam shall be of corrosion resistant steel. 5.0 gram and 0.5 gram weights shall be of nickel-copper-alloy wire and the 0.05 gram and 0.005 gram weights shall be of aluminum wire.

The forceps shall be of stainless steel.

Knife edges and bearings shall be of agate.

DESIGN: Balance shall be of the Westphal design and shall consist of the following component parts: hydrometer jar, counterpoise weight, stirrup and beam, a thermometer plummet, a pillar, rider weights, knives and beams and shall be (A) assembled in a padded box. The balance shall have an accuracy of porm 0.1 mg.

(A) Hydrometer jar shall be a cylinder 5 inches (127 mm) high.

(A) A glass hydrometer-plummet shall have an overall length of 105 mm and an outer diameter 10 mm wide at its greatest width, and include a brass hook and a 5 inch (127 mm) length of platinum wire to attach the thermometer to the balance beam. Plummet weight shall be of 11.75 grams porm 0.25 grams. It shall have a displacement of 5.000 mL of water porm 0.003 mL at 15°C.

The beam shall be equipped with 9 evenly spaced notches for rider weight suspension, numbered from 1 to 9, and have 2 knife edges ground and polished. Rider weights (8) shall be of the hairpin type designed for the Westphal balance.

APPROVED 7 OCT 1968 REVISED (A) 24 JUL 1986

P.A. EA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust MS 99		BALANCE, SPECIFIC GRAVITY DETERMINATION, WESTPHAL	MS36537
Procurement Specifications NONE	SUPERSEDES:		SHEET 1 OF 2

PROJECT NUMBER 6630-0437-4

DD FORM 1 MAY 73 672 (LIMITED COORDINATION) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:42Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS
6630

Base shall be so constructed as to prevent tilting when in use, and shall be provided with a manually adjustable leveling screw suitable for leveling the balance.

A forceps with curved tips, 100 mm long shall be included with the set.

PURPOSE: For accurate specific gravity determinations of liquids to the 4th decimal place by measuring weight loss of an immersed plummet of standard weight and volume.

NOTES:
1. Items differing from this Military Standard by minor variations in design or dimensions are considered standard under this document, provided such variations do not affect performance and are not in conflict with the material requirements as specified herein.
2. Referenced documents shall be of the issue in effect on date of invitations for bid.
Ⓐ 3. FSCMs: 20512, 21519, and 22527.

USER ACTIVITIES:

REVIEWER ACTIVITIES:

REVISED

APPROVED

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. EA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust MS 99		BALANCE, SPECIFIC GRAVITY DETERMINATION, WESTPHAL	MS36537
Procurement Specifications NONE	SUPERSEDES:		SHEET 2 OF 2

DD FORM 1 MAY 73 672 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

PROJECT NUMBER 6630-0437-4
PROPOSED REVISION DATE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:42Z
Check the source to verify that this is the current version before use.